

Jason D. Hartline

Professor of Computer Science
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Research Interests

 **Economics.** Mechanism design, microeconomics, econometrics, economics of algorithms.

 **Law.** Algorithmic law, accountability, regulating algorithms, science of law.

 **Artificial Intelligence.** Explainability, fairness, alignment, future of work.

 **Data Science.** Machine learning, outcomes from learning agents, regulating algorithms.

Education

Ph.D. in Computer Science. University of Washington, Seattle, WA. *Summer 2003*
Thesis: Optimization in the Private Value Model: Competitive Analysis Applied to Auction Design
Advisor: Anna Karlin.

M.S. in Computer Science. University of Washington, Seattle, WA. *Spring 2000*

B.S. in Computer Science. Cornell University, Ithaca, NY. *Spring 1997*

B.S. in Electrical Engineering. Cornell University, Ithaca, NY. *Spring 1997*

Current Appointment

Professor. Northwestern U., Evanston, IL. *Fall 2019 – present*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Cofounder. Virtual Chair Inc., Chicago IL. *Summer 2020 – present*

Previous Appointments

Visiting Professor. Stanford U., Stanford, CA. *2023-2024*
Economics department and Graduate School of Business

Associate Professor. Northwestern U., Evanston, IL. *Fall 2012 – Summer 2019*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Visiting Researcher. Microsoft Research, Cambridge, MA. *Spring 2015*

Visiting Professor. Harvard U., Cambridge, MA. *2014*
Computer Science and Economics departments.

Assistant Professor. Northwestern U., Evanston, IL. *Winter 2008 – Summer 2012*
 Electrical Engineering and Computer Science Department, McCormick School of Engineering and
 Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy).

Researcher. Microsoft Research, Mountain View, CA. *2004 – 2007*
 Research Area: Algorithmic Mechanism Design, Auction Theory, Pricing Algorithms, Auctions for
 Sponsored Search.

Post-doctoral Research Fellow. ALADDIN, Carnegie Mellon U., Pittsburgh, PA. *Fall 2003*
 Research Area: Mechanism Design.
 Supervisor: Avrim Blum.

Mentoring

Ph.D. Advisees. *current*
 Anant Shah, Matthew vonAllmen, Chang Wang, Yifan Wu, Chenhao Zhang.

Former Students. *since 2009*
 Modibo Camara (Stanford), Yingkai Li (Yale), Yiding Feng (Microsoft Research), Bach Ha (Mi-
 crosoft), Aleck Johnsen, Nima Haghpahan (Penn State, Asst. Prof.), Darrell Hoy (Carta), Michalis
 Mamakos (Northwestern, postdoc), and Samuel Taggart (Oberlin, Asst. Prof.)

Post-doctoral Fellows. *since 2006*
 Jinshuo Dong, Hedyeh Beyhaghi (CMU), Liad Blumrosen (Hebrew U.), Azarakhsh Malekian
 (Toronto)

Short-term Students. *since 2004*
 Gagan Aggarwal, Abraham Flaxman, Ning Chen, Mukund Sundararajan, Benjamin Prosnitz,
 Matthew Burgess, Saeed Alaei, Hu Fu, Shweta Jain, Rad Niazadeh, Sarah Lim, Sadie Hood.

Service

 **Co-organizer.** Monthly CS+Law Workshop *2021 – present*
<https://www.cslawworkshop.org/>

 **Program Committee.** ACM Conf. on Economics & Computation. *2005, 2006, 2008 – present*

  **Program Committee.** Symposium on Foundations of Responsible Computing *2024*

 **Program Chair.** ACM Conference on Economics and Computation. *2023*

 **Program Committee.** ACM Symposium on CS and Law. *2023*

 **Associate Editor.** Journal of Economic Theory *2019 – present*

  **Co-organizer.** IDEAL Special Quarter on Data Economics. *Fall 2022*

 **Co-organizer.** IDEAL Special Quarter on CS and Law. *Spring 2021*

 **Virtual Chair.** ACM Conference on Economics and Computation *2020*

 **Co-organizer.** Northwestern CS+Econ Quarterly Workshop *2018 – 2020*

Co-organizer. Northwestern Quarterly Theory Workshop *2016 – 2020*

 **General Chair.** ACM Conference on Electronic Commerce. *2020*

☞	☞	Co-organizer. Special Quarter on Online Markets and Data Science with Jacob Abernethy, Constantinos Daskalakis, and Denis Nekipelov.	<i>Spring 2018</i>
☞		Special Initiatives Chair. ACM Special Interest Group on E-commerce. on the Academic Job Market.	<i>2014 – 2015</i>
☞		Guest Editor. Games and Economic Behavior. special issues for papers from STOC, FOCS, and SODA conferences.	<i>2011 – 2014</i>
☞		Advisory Editor. Games and Economic Behavior.	<i>2012 – 2017</i>
		Associate Editor. Operations Research Letters.	<i>2011 – 2014</i>
☞		Co-organizer. New York Computer science and Economics (NYCE) Day.	<i>2013</i>
☞		Co-organizer. FOCS Workshop on Bayesian Mechanism Design.	<i>2014</i>
		Program Committee. Symposium on Theory of Computation	<i>2012</i>
☞		Co-organizer. Workshop on Bayesian Mechanism Design	<i>2011</i>
☞		Co-organizer. Greece Economic and Algorithmic Theory Week.	<i>2011, 2014</i>
☞		Co-organizer. Bertinoro Workshop on Algorithmic Game Theory.	<i>2006, 2010</i>
☞		Tutorials Chair. ACM Conference on Electronic Commerce.	<i>2010</i>
☞		Local Arrangements. ACM Conference on Electronic Commerce.	<i>2008</i>
		Organizer. Midwest Theory Day.	<i>2008</i>
		Program Committee. ACM-SIAM Symposium on Discrete Algorithms.	<i>2007</i>
☞		Co-organizer. Bay Algorithmic Game Theory Symposium (biannual).	<i>2006 – 2007</i>
☞		Co-organizer. Workshop on Sponsored Search Auctions.	<i>2006</i>
☞		Co-organizer. Alternative Solution Concepts in Mechanism Design.	<i>2006</i>
☞		Co-organizer. ALADDIN Workshop on Auction Theory & Practice.	<i>2003</i>

Awards, Fellowships, and Grants

		Ver Steeg Award. Northwestern U. for graduate student mentorship	<i>2025</i>
☞	🎓	FORC Best Student Paper. with Jiale Chen and Onno Zoeter for “Fair Grading of Randomized Exams”	<i>2023</i>
☞		NSF AF. Mechanism Design for the Classroom	<i>2022</i>
☞		ESA Test of Time. with Andrew Goldberg for “Competitive Auctions and Multiple Digital Goods” from ESA 2001.	<i>2021</i>
☞		SIGecom Test of Time. with Andrew Goldberg and Andrew Wright for “Competitive Auctions and Digital Goods” from SODA 2021.	<i>2021</i>

-  **NSF TRIPODS.** Institute for Data, Econometrics, Algorithms and Learning
co-dorector; with Aravindan Vijayaraghavan and others. 2019
-  **NSF AitF.** Mechanism Design and Machine Learning for Peer Grading
with Douglas Downey and Eleanor O'Rourke. 2017
-  **NSF AF.** Non-revelation Mechanism Design 2016
- Teacher of the Year.** EECS Dept., Northwestern U. 2011
-   **NSF ICES.** Towards Realistic Mechanisms: statistics, inference, and approximation in simple
Bayes-Nash implementation. with Shuchi Chawla and Denis Nekipelov. 2011
-  **NSF CAREER Award.** Mechanism Design 2009
-  **NSF TF.** Mechanism Design and Approximation 2008
with Shuchi Chawla.
- ALADDIN Post-doctoral Research Fellowship.** Carnegie Mellon University. 2003
- NSF Math Sciences Post-doctoral Research Fellowship.** Declined. 2003
- Microsoft Endowed Fellowship.** CS Dept., U. of Washington 2001
- Bob Bandes Teaching Award, Honorable Mention.** CS Dept., U. of Washington 1998
- Small Business Innovative Research Grant.** Department of Education 1997

Patents

-  **Online Pricing and Buyback.** U.S. Patent #8260724 2012
with Moshe Babaioff and Robert Kleinberg.
-  **Systems and Methods for Pricing and Selling Digital Goods.** U.S. Patent #6985885 2006
with Andrew Goldberg and Andrew Wright.

Book Chapters

-  **Profit Maximizing Mechanism Design.** Algorithmic Game Theory 2007
with Anna Karlin; eds. Noam Nisan, Tim Roughgarden, Eva Tardos, and Vijay Vazirani.

Popular Press

-  **Badminton and the Science of Rule Making.** *Huffington Post* 2012
with Robert Kleinberg.

Working Papers

-   **ElicitationGPT: Text Elicitation Mechanisms via Language Models.** 2024
with Yifan Wu.
- Subgame Optimal and Prior-Independent Online Algorithms.** 2024
with Aleck Johnsen and Anant Shah.

▣ **Algorithmic robust forecast aggregation.** 2024
with Yongkang Guo, Zhihuan Huang, Yuqing Kong, Anant Shah, and Fang-Yi Yu.

▣ **Mechanism Redesign.** 2017
with Shuchi Chawla and Denis Nekipelov.

Journal Papers

▣ **Fast Core Pricing for Rich Advertising Auctions.** *Operations Research* 2022
with Rad Niazadeh, Nicole Immorlica, Mohammad Resa Khani, and Brendan Lucier.

▣ **When is pure bundling optimal?.** with Nima Haghpahan *Review of Economic Studies* 2021

▣ **Bernoulli factories and black-box reductions in mechanism design.** *JACM* 2021
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh.

▣ **Efficient Computation of Optimal Auctions via Reduced Forms.** *Math of OR* 2019
with Saeed Alaei, Hu Fu, Nima Haghpahan, and Azarakhsh Malekian.

▣ **Optimal auctions vs. Anonymous Pricing.** *Games and Economic Behavior* 2018
with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan. Special issue.

▣ **Non-optimal Mechanism Design.** *American Economic Review* 2015
with Brendan Lucier.

▣ **Bayesian Incentive Compatibility and Matchings.** *Games and Economic Behavior* 2015
with Robert Kleinberg and Azarakhsh Malekian. Special issue.

▣ **Mechanism Design via Consensus Estimates, Cross Checking, and Profit Extraction.**
with Bach Ha. Special issue. *Transactions on Economics and Computation* 2013

▣ **Optimal Crowdsourcing Contests.** *Games and Economic Behavior* 2015
with Shuchi Chawla and Balu Sivan. Special issue.

▣ **Envy freedom and prior-free mechanism design.** *Journal of Economic Theory* 2015
with Nikhil Devanur and Qiqi Yan. Special issue.

▣ **Bayesian Mechanism Design.** *FTTCS*¹ 2012

▣ **Approximation in Mechanism Design.** *American Economic Review* 2012

▣ **Derandomization of Auctions.** *Games and Economic Behavior* 2010
with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.

▣ **Algorithms for Data Migration.** *Algorithmica* 2010
with Eric Anderson, Joseph Hall, M. Hobbes, Anna Karlin, Jared Saia, Ram Swaminathan, and John Wilkes.

▣ **Reducing Mechanism Design to Algorithm Design via Machine Learning.** *JCSS*² 2008
with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.

▣ **Competitive Auctions.** *Games and Economic Behavior* 2006
with Andrew Goldberg, Anna Karlin, Mike Saks, and Andrew Wright. Special issue.

¹Foundations and Trends in Theoretical Computer Science.

²Journal of Computer and Systems Sciences.

Characterizing History Independent Data Structures.*Algorithmica* 2005

with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke. Special issue.

Refereed Conference Papers

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 **Smooth Calibration and Decision Making.** *FORC*³ 2025
 with Yifan Wu and Yunran Yang
- 
 **Underspecified Human Decision Experiments Considered Harmful.** *CHI*⁴ 2025
 with Jessica Hullman and Alex Kale
- 

 **Regulation of Algorithmic Collusion, Refined: Testing Pessimistic Calibrated Regret.** *CSLaw*⁵ 2025
 with Chang Wang and Chenhao Zhang
-  **Combinatorial Pen Testing (or Consumer Surplus of Deferred-Acceptance Auctions).** *ITCS*⁶ 2025
 with Aadityan Ganesh
-  **Fundamental Limits of Throughput and Availability: Applications to prophet inequalities and transaction fee mechanism design.** *EC*⁷ 2024
 with Adityan Ganesh, Atanu Sinha, and Matthew vonAllmen
-  **A Decision Theoretic Framework for Measuring AI Reliance.** *FAccT*⁸ 2024
 with Ziyang Guo, Yifan Wu, and Jessica Hullman.
- 
 **Designing Shared Info Displays for Agents of Varying Strategic Sophist.** *CSCW*⁹ 2024
 with Dongping Zhang and Jessica Hullman.
- 
 **Error-Tolerant E-Discovery Protocols.** *CSLaw* 2024
 with Jinshuo Dong, Liren Shan, and Aravindan Vijayaraghavan.
- 

 **Regulation of algorithmic collusion.** with Sheng Long and Chenhao Zhang *CSLaw* 2024
-  **Equivocal Blends: Prior Independent Lower Bounds.** *ITCS* 2024
 with Aleck Johnsen
-  **The Rational Agent Benchmark for Data Visualization.** *VIS*¹⁰ 2023
 with Yifan Wu, Ziyang Guo, Michalis Mamakos, Jessica Hullman
-  **Optimal Scoring Rules for Multi-dimensional Effort.** *COLT*¹¹ 2023
 with Liren Shan, Yingkai Li, and Yifan Wu
- 
 **Fair Grading Algorithms for Randomized Exams.** *FORC* 2023
 with Jiale Chen and Onno Zoeter. Best student paper award.

³Symposium on Foundations of Responsible Computing.⁴ACM CHI Conference on Human Factors in Computing Systems⁵ACM Symposium on Computer Science and Law⁶Innovations in Theoretical Computer Science Conference⁷ACM Conference on Economics and Computation⁸ACM Conference on Fairness, Accountability, and Transparency⁹ACM Conference on Computer-Supported Cooperative Work and Social Computing¹⁰IEEE Transactions on Visualization and Computer Graphics¹¹Conference on Learning Theory.

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Screening with Disadvantaged Agents. *FORC 2023*
 with Hedyeh Beyhaghi, Modibo Camara, Aleck Johnsen, and Sheng Long
- 


Non-strategic Econometrics (for Initial Play). *AAMAS¹² 2023*
 with Daniel Chui and James Wright
- 
Simple Mechanisms for Non-linear Agents. *SODA¹³ 2023*
 with Yiding Feng and Yingkai Li
- 

Algorithmic Learning Foundations for Common Law. *CSLaw 2022*
 with Dan Linna, Liren Shan, and Alex Tang
- 


Classification Protocols with Minimal Disclosure. *CSLaw 2022*
 with Jinshuo Dong and Aravindan Vijayaraghavan
- Karp: A Language for NP Reductions.** *PLDI¹⁴ 2022*
 with Chenhao Zhang and Christos Dimoulas
- 
Visualization Equilibrium. *IEEE VIS¹⁵ 2021*
 with Paula Kayongo, Glen Sun, and Jessica Hullman
- 
Revelation Gap for Pricing from Samples. with Yiding Feng and Yingkai Li *STOC¹⁶ 2021*
- 
Welfare-maximizing Guaranteed Dashboard Mechanisms. *EC 2021*
 with Yuan Deng, Jieming Mao, and Balasubramanian Sivan
- 
Non-quasi-linear Agents in Quasi-linear Mechanisms. *ITCS 2021*
 with Moshe Babaioff, Richard Cole, Nicole Immorlica, and Brendan Lucier
- 

Mechanisms for a no-regret agent: Beyond the common prior. *FOCS¹⁷ 2020*
 with Modibo Camara and Aleck Johnsen
- 
Benchmark design and prior-independent optimization. *FOCS 2020*
 with Aleck Johnsen and Yingkai Li
- 

Inference from Prices. *SODA 2020*
 with Aleck Johnsen, Denis Nekipelov, and Zihe Wang.
- 
A Truthful Cardinal Mechanism for One-Sided Matching. *SODA 2020*
 with Rediet Abebe, Richard Cole, and Vasilis Gkatzelis.
- 

Sample Complexity for Non-Truthful Mechanisms. *EC 2019*
 with Samuel Taggart.
- 
Dashboard Mechanisms for Online Marketplaces. *EC 2019*
 with Aleck Johnsen, Denis Nekipelov, and Onno Zoeter.
- 
Optimal Auctions vs. Anonymous Pricing: Beyond Linear Utility. *EC 2019*
 with Yiding Feng and Yingkai Li.

¹²International Conference on Autonomous Agents and Multiagent Systems

¹³ACM-SIAM Symposium on Discrete Algorithms.

¹⁴ACM SIGPLAN International Conference on Programming Language Design and Implementation

¹⁵IEEE Transactions on Visualization and Computer Graphics

¹⁶ACM Symposium on Theory of Computing.

¹⁷IEEE Symposium on Foundations of Computer Science.

- An End-to-end Argument in Mechanism Design (Prior-independent Auctions for Budgeted Agents).** with Yiding Feng. *FOCS 2018*
- Fast Core Pricing for Rich Advertising Auctions.** *EC 2018*
with Nicole Immorlica, Mohammad Reza Khani, Brendan Lucier, and Rad Niazadeh.
- Bernoulli Factories and Black-box Reductions in Mechanism Design.** *STOC 2017*
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh
- Bayesian Budget Feasibility with Posted Pricing.** *WWW¹⁸ 2016*
with Eric Balkanski.
- A/B Testing of Auctions.** *EC 2016*
with Shuchi Chawla and Denis Nekipelov.
- No-regret Learning in Bayesian Games.** *NeurIPS¹⁹ 2015*
with Vasilis Syrgkanis and Eva Tardos.
- Reverse Mechanism Design.** *EC 2015*
with Nima Haghpanah.
- Optimal auctions vs. Anonymous Pricing.** *FOCS 2015*
with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan.
- Mechanism Design for Data Science.** *EC 2014*
with Shuchi Chawla and Denis Nekipelov.
- Price of Anarchy for Auction Revenue.** *EC 2014*
with Darrell Hoy and Samuel Taggart.
- Optimal Auctions for Correlated Buyers with Sampling.** *EC 2014*
with Hu Fu, Nima Haghpanah, and Robert Kleinberg.
- The Simple Economics of Approximately Optimal Auctions.** *FOCS 2013*
with Saeed Alaei, Hu Fu, and Nima Haghpanah.
- Auctions with Unique Equilibria.** *EC 2013*
with Shuchi Chawla.
- Prior-independent Auctions for Risk-averse Agents.** *EC 2013*
with Hu Fu and Darrell Hoy.
- Prior-free Auctions for Budgeted Agents.** *EC 2013*
with Nikhil Devanur and Bach Ha.
- Prior-independent Mechanisms for Scheduling.** *STOC 2013*
with Shuchi Chawla, David Malec, and Balu Sivan.
- Mechanism Design via Multi- to Single-agent Reduction.** *EC 2012*
with Saeed Alaei, Hu Fu, Nima Haghpanah, and Azarakhsh Malekian.

¹⁸International Conference on the World Wide Web.¹⁹Conference on Neural Information Processing Systems

- Optimal Crowdsourcing Contests.** *SODA 2012*
 with Shuchi Chawla and Balu Sivan. Invited to GEB special issue.
- Mechanism Design via Consensus Estimates, and Cross Checking, and Profit Extraction.** With Bach Ha. Invited to TEAC special issue. *SODA 2012*
- Truth, Envy, and Profit.** *EC 2011*
 with Qiqi Yan. Invited to JET special issue.
- Bayesian Incentive Compatibility and Matchings.** *SODA 2011*
 with Robert Kleinberg and Azarakhsh Malekian. Invited to GEB special issue.
- Bayesian Algorithmic Mechanism Design.** *STOC 2010*
 with Brendan Lucier.
- Sequential Posted Pricing and Multi-parameter Mechanism Design.** *STOC 2010*
 with Shuchi Chawla, David Malec, and Balasubramanian Sivan.
- Simple versus Optimal Mechanisms.** *EC 2009*
 with Tim Roughgarden.
- Limited and Online Supply and the Bayesian Foundations of Prior-free Mechanism Design.** with Nikhil Devanur. *EC 2009*
- Selling Ad Campaigns: Online Algorithms with Cancellations.** *EC 2009*
 with Moshe Babaioff and Robert Kleinberg.
- Mechanism Design and Money Burning.** *STOC 2008*
 with Tim Roughgarden.
- Optimal Marketing Strategies over Social Networks.** *WWW 2008*
 with Vahab Mirrokni and Mukund Sundararajan.
- Auctions for Structured Procurement.** *SODA 2008*
 with Matthew Cary, Abraham Flaxman, and Anna Karlin.
- Algorithmic Pricing via Virtual Valuations.** *EC 2007*
 with Shuchi Chawla and Robert Kleinberg.
- Knapsack Auctions.** *2006*
 with Gagan Aggarwal.
- Bayesian Optimal No-deficit Mechanism Design.** *WINE*²⁰ *2006*
 with Shuchi Chawla, R. Ravi, and Uday Rajan.
- Mechanism Design via Machine Learning.** *FOCS 2005*
 with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.
- Derandomization of Auctions.** *STOC 2005*
 with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.
- On Profit-Maximizing Envy-Free Pricing.** *SODA 2005*
 with Venkat Guruswami, Anna Karlin, David Kempe, Claire Kenyon, and Frank McSherry.

²⁰Conference on Web and Internet Economics.

- Collusion-Resistant Mechanisms for Single Parameter Agents.** *SODA 2005*
 with Andrew Goldberg.
- Near-Optimal Online Auctions.** *SODA 2005*
 with Avrim Blum.
- From Optimal Limited to Unlimited Supply Auctions.** *EC 2005*
 with Robert McGrew.
- On the Competitive Ratio of the Random Sampling Auction.** *WINE 2005*
 with Uriel Feige, Abraham Flaxman, and Robert Kleinberg.
- Near-Optimal Pricing in Near-Linear Time.** *WADS²¹ 2005*
 with Vladlen Koltun.
- A Lower Bound on the Competitive Ratio of Truthful Auctions.** *STACS²² 2004*
 with Andrew Goldberg, Anna Karlin, and Mike Saks.
- Competitiveness via Consensus.** *SODA 2003*
 with Andrew Goldberg
- Envy-Free Auctions for Digital Goods.** *EC 2003*
 with Andrew Goldberg.
- Truthful and Competitive Double Auctions.** *ESA²³ 2002*
 with Kaustubh Deshmukh, Andrew Goldberg, and Anna Karlin.
- Competitive Generalized Auctions.** *STOC 2002*
 with Amos Fiat, Andrew Goldberg, and Anna Karlin.
- Characterizing History Independent Data Structures.** *ISAAC²⁴ 2002*
 with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke.
- Competitive Auctions for Multiple Digital Goods.** *ESA 2001*
 with Andrew Goldberg.
- Competitive Auctions and Digital Goods.** *SODA 2001*
 with Andrew Goldberg and Andrew Wright.
- On Algorithms for Efficient Data Migration.** *SODA 2001*
 with Joe Hall, Anna Karlin, Jared Saia, and John Wilkes.
- An Experimental Study of Data Migration Algorithms.** *WAE²⁵ 2001*
 with E. Anderson, J. Hall, M. Hobbess, A. Karlin, J. Saia, R. Swaminathan, and J. Wilkes.

²¹Workshop on Algorithms and Data Structures.

²²Symposium on Theoretical Aspects of Computer Science.

²³European Symposium on Algorithms.

²⁴International Symposium on Algorithms and Computation.

²⁵International Workshop on Algorithm Engineering.